

MAISHA RAHMAN CHOWDHURY

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EDUCATION

Iowa State University, Ames, IA

Aug 2022 – May 2026

Bachelor of Science in Computer Science, Minor in Applied Artificial Intelligence

GPA: 3.72 | Magna Cum Laude

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, Bash

Backend: Node.js, Spring Boot, REST APIs, WebSockets

Databases: MySQL, MongoDB, Neo4j, Firebase

AI/Systems: Linux, Raspberry Pi, LLMs, LangGraph, Docker, LinUCB, Machine Learning

Tools: Git, GitLab CI/CD, Postman, Jupyter Notebook, Conda, LM Studio

RESEARCH EXPERIENCE

Adaptive Agentic Scheduling for Multi-Model LLM Inference | Dr. Asad W. Malik

May 2026 – Present

- Extend iGenOrch with LinUCB-based adaptive scheduling for concurrent LLM inference.
- Use runtime telemetry to optimize CPU allocation, latency, and token budgets under workload contention.
- Integrate LangGraph to coordinate telemetry, scheduling decisions, and reward updates.

Undergraduate Research Fellow – Summer iEDGE REU | Mississippi State University

May 2025 – Aug 2025

- Developed a Linux kernel-aware scheduler in C on Raspberry Pi 5 to prioritize LLM inference workloads.
- Deployed TinyLLaMA and DeepSeek on Linux edge systems and automated experiments with Python/Bash.
- Collected CPU, memory, and latency telemetry to evaluate scheduling trade-offs for edge AI workloads.

Undergraduate Research Assistant | Iowa State University

Jun 2024 – Aug 2024

- Researched interpretability and reliability of code-trained LLMs under Prof. Janessari.
- Built a Python/Tkinter annotation tool to compare LLM-generated and human labels.
- Analyzed annotation discrepancies to evaluate LLM labeling reliability.

PUBLICATION

Akram, F., Chowdhury, M. R., et al. “iGenOrch: Intelligent Orchestration Framework for Multi-Model LLM Inference on Edge Platforms.” Proceedings of ACMSE 2026.

PROJECTS

Throwly – Software Developer, Client-Sponsored Computer Science Capstone

Jan 2026 – May 2026

- Developed React Native/TypeScript features and Node.js backend workflows for a student marketplace.
- Integrated Didit identity verification with frontend WebView and backend validation.
- Implemented reverse geocoding and resolved navigation issues across five core screens.

Clever Quest – Real-Time Multiplayer Game Platform

Aug 2024 – Dec 2024

- Built a Spring Boot/Java multiplayer backend using WebSockets for real-time game communication.
- Designed REST APIs for authentication, game sessions, accounts, and score persistence.
- Designed SQL schemas and queries for game state, accounts, and leaderboards; used GitLab CI/CD for collaborative development.

Dungeon Crawler Game Engine

Jan 2025 – May 2025

- Built a C/C++ terminal-based dungeon crawler with manual memory management.
- Implemented procedural dungeon generation and grid-based navigation.
- Developed AI behaviors using pathfinding, priority queues, and randomized movement.

WORK EXPERIENCE

Teaching Assistant | Department of Computer Science, Iowa State University

Aug 2024 – Dec 2024

- Led Python lab sessions and office hours for Com S 127, supported students with programming concepts.
- Graded assignments and collaborated with faculty on exam rubrics and student feedback.